

Given a map $f: \mathbb{R} \rightarrow \mathbb{R}$, a map

$$g: x \mapsto \int_a^b K(x, t) f(t) dt$$

is called the 'integral transform'.

In this page, we cover typical int. forms.

Laplace transform.

Given a map $f: \mathbb{R} \rightarrow \mathbb{C}$, a Laplace transform of f is given by:

$$\mathcal{L}f: \mathbb{C} \rightarrow \mathbb{C}; x \mapsto \int_0^{\infty} e^{-xt} \cdot f(t) dt.$$

Prop. If f is differentiable and $e^{-xR} \cdot f(R) \rightarrow 0$ as $R \rightarrow \infty$,

$$\text{then } (\mathcal{L}f)'(x) = \int_0^{\infty} e^{-xt} \cdot f'(t) dt = \lim_{R \rightarrow \infty} \left[e^{-xt} f(t) \Big|_0^R \right] + x \cdot \int_0^{\infty} e^{-xt} \cdot f(t) dt$$

$$= x \cdot (\mathcal{L}f)(x) - f(0)$$

Fourier transform.

Convolution

Given two maps $f, g: \mathbb{R} \rightarrow \mathbb{C}$, define

$$(f * g)(x) = \int_{-\infty}^{\infty} g(x-t) f(t) dt \quad x-t \in [-r, r] \Leftrightarrow -r+x$$

Prop. Let $k_r(x) = \frac{1}{2r} \cdot \chi_{[-r, r]}(x)$. Then,

$$(f * k_r)(x) = \int_{-\infty}^{\infty} k_r(x-t) \cdot f(t) dt = \frac{1}{2r} \cdot \int_{x-r}^{x+r} f(t) dt \quad (1)$$

Moreover, $\int_{-\infty}^{\infty} |f| < \infty \Rightarrow f * k_r$ is uniformly continuous. (2)

and f is uniformly continuous $\Rightarrow f * k_r \rightarrow f$ as $r \rightarrow 0$ uniformly. (3)

Proof. (1) can be shown by:

$$g(x-t) \cdot f(t) = 0 \Leftrightarrow x-t \notin [-r, r] \Leftrightarrow x-r \leq t \leq x+r.$$

To show (2), let $r > 0$ be fixed. Given $\epsilon > 0$, take $\delta > 0$ such that

$$0 < t-s < \delta \Rightarrow \int_s^t |f| < r\epsilon.$$

Now, put $\delta = \min\{2r, \delta_0\}$. Then, $|x-y| < \delta$ implies

$$\begin{aligned} |(f * k_r)(x) - (f * k_r)(y)| &= \frac{1}{2r} \cdot \left| \int_{x-r}^{x+r} f(t) dt - \int_{y-r}^{y+r} f(t) dt \right| \quad |x-y| < \delta < 2r. \\ &= \frac{1}{2r} \cdot \left| \int_{x-r}^{y-r} f(t) dt - \int_{x+r}^{y+r} f(t) dt \right| \\ &< \frac{1}{2r} \cdot (r\epsilon + r\epsilon) = \epsilon. \end{aligned}$$

To show (3), let $\epsilon > 0$ be given. Then, there exists $\delta > 0$ such that

$$|x-y| < \delta \Rightarrow |f(x) - f(y)| < \epsilon.$$

Now, for $r < \delta$,

$$\begin{aligned} |f(x) - (f * k_r)(x)| &= \left| \frac{1}{2r} \cdot \int_{x-r}^{x+r} f(t) dt - f(x) \right| \\ &\leq \frac{1}{2r} \cdot \int_{x-r}^{x+r} |f(x) - f(t)| dt < \epsilon. \end{aligned}$$

Therefore, $f * k_r \rightarrow f$ uniformly. $\square$